

M303

TMA 05

2025J

Mainly covers Book E, *Rings and fields*

Cut-off date 31 March 2026

You will find instructions for completing TMAs in the Assessment resources area of the M303 website. Please remind yourself of these instructions before beginning work on each TMA. You can submit each TMA electronically by using the University's online TMA/EMA service or by post together with completed TMA form (PT3). In both cases the University must receive it by the cut-off date given above unless you have agreed an extension in advance with your tutor.

Half of each TMA is formative and the other half summative. The formative questions are designed to *extend* your understanding of the topics, while the summative ones place more emphasis on *assessing* your understanding of the topics. Both types of question will be marked by your tutor, but (unlike the summative questions) marks for the formative questions do not count towards your final grade. You should submit your solutions to both sets of questions to your tutor by the cut-off date.

The marks allocated to each part of a question are indicated in the margin.

Although many of the questions on these assignments require numerical answers, the questions invariably carry *method marks*.

The words listed below, when used in questions, should be interpreted as indicated.

prove, show, explain, justify	Clear reasoning and explanation for all steps are called for.
determine, find, devise, calculate, compute	An indication of the method used and all working in arriving at an answer should be given.
deduce	Clear explanation of how one result follows from another is required.
solve	Working must be shown. A numerical answer alone is not sufficient.
evaluate, give, write down, list	Answer suffices. No explanation need be given.
hence	No marks will be awarded for any alternative method.

PLAGIARISM WARNING – the use of assessment help services and websites

The work that you submit for any assessment/examination on any module should **be your own**. Submitting work produced by or with another person, or a web service or an automated system, **as if it is your own** is cheating. It is **strictly forbidden** by the University.

You should not:

- provide any assessment question to a website, online service, social media platform or any individual or organisation, as this is an infringement of copyright
- request answers or solutions to an assessment question on any website, via an online service or social media platform, or from any individual or organisation
- use an automated system (other than one prescribed by the module) to obtain answers or solutions to an assessment question and submit the output as your own work
- discuss examination questions with any other person, including your tutor.

The University actively monitors websites, online services and social media platforms for answers and solutions to assessment questions, and for assessment questions posted by students. Work submitted by students for assessment is also monitored for plagiarism.

A student who is found to have posted a question or answer to a website, online service or social media platform and/or to have used any resulting, or otherwise obtained, output as if it is their own work has committed a disciplinary offence under our [Code of Practice for Student Discipline](#). **This means the academic reputation and integrity of the University has been undermined.**

The Open University's [Academic Conduct Policy](#) defines plagiarism in part as:

- using text obtained from assignment writing sites, organisations or private individuals
- obtaining work from other sources and submitting it as your own.

If it is found that you have used the services of a website, online service or social media platform, or that you have otherwise obtained the work you submit from another person, this is considered serious academic misconduct and you will be referred to the Central Disciplinary Committee for investigation.

This TMA is based on Book E (Rings and fields) except for the formative Question 8.

Part A, consisting of Questions 1 – 5, 6A, 6B and 7, worth a total of 50 marks, is **summative**. Full marks may be obtained by completing correctly Questions 1 – 5 and 7, and **one** of 6A, 6B. Please only submit one of 6A and 6B. If you submit both parts then your tutor will only mark the first part submitted. (Providing you submit one solution to Q6 you will receive both feedback on your attempt and the solution to the other part of Q6 when your TMA is returned.)

These questions assess your knowledge of material in the text. You must not discuss these questions in the forums, and your tutor is not allowed to help you directly.

Questions 1, 3 and 4 of Part A are designed to demonstrate understanding of definitions and basic concepts and to practise calculations in various rings and fields. Questions 2 and 5 include some revision of earlier material.

Part B is formative. You should choose at most two formative questions to submit to your tutor for marking.

Doing the formative questions will help you to understand key concepts covered in the text. You are encouraged to discuss these questions in the module forums and with your tutor (who is allowed to help you).

Your tutor will mark all the summative answers that you submit and the first two of any formative answers that you submit. If you have attempted at least one formative question then when your TMA is returned you will be sent worked solutions to all the formative questions. Note that only your marks for Part A count towards your overall continuous assessment score.

Part A Summative

Question 1 – 9 marks

This M303 questions enables you to practise working with ideals in rings that are extensions of degree two over the integers. It involves a number of concepts from Chapter 17.

Let R be the ring $\mathbb{Z}[\sqrt{-14}]$.

(a) Let A be the subset of R defined by

$$A = \{a + b\sqrt{-14} : a, b \in \mathbb{Z}, a - b \equiv 0 \pmod{5}\}.$$

Show that A is an ideal in the ring R .

[5]

(b) The subset

$$B = \{c + d\sqrt{-14} : c, d \in \mathbb{Z}, c + d \equiv 0 \pmod{5}\}$$

is also an ideal in the ring R (you do not need to prove this).

Show that $AB = \langle 5 \rangle$.

[4]

Question 2 – 4 marks

This M303 question revises material from Chapter 11 and uses concepts and results from Chapters 17, 18 and 19.

- (a) Show that $f(x) = 3x^4 + 156x^3 + 26x^2 + 130x + 26 \in \mathbb{Z}[x]$ is irreducible over $\mathbb{Q}$. [1]
- (b) Let $\alpha \in \mathbb{C}$ and $\beta \in \mathbb{C}$ be two distinct roots of $f(x)$. Is $\mathbb{Q}(\alpha) \cong \mathbb{Q}(\beta)$? Briefly justify your answer. [3]

Question 3 – 7 marks

For this M303 question you will need to use concepts and results from Chapter 18.

Let n be an odd positive integer that is not a perfect square and that is not equal to 0 (mod 13). You may assume that $\sqrt{13} \notin \mathbb{Q}$ and that $x^2 - 13 \in \mathbb{Q}[x]$ is irreducible over $\mathbb{Q}$.

Let $\alpha = \sqrt{13} + \sqrt{n}$.

- (a) By evaluating $\alpha^3 - (n + 39)\alpha$, show that $\mathbb{Q}(\sqrt{13}, \sqrt{n}) \subseteq \mathbb{Q}(\sqrt{13} + \sqrt{n})$. [3]
- (b) Deduce that $\mathbb{Q}(\sqrt{13}, \sqrt{n}) = \mathbb{Q}(\sqrt{13} + \sqrt{n})$. [1]
- (c) Determine the *minimal* polynomial for $\alpha = \sqrt{13} + \sqrt{n}$ over $\mathbb{Q}$. (You may assume that $[\mathbb{Q}(\sqrt{13}, \sqrt{n}) : \mathbb{Q}] = 4$). [3]
- Hint: start by evaluating α^2 .

Question 4 – 9 marks

This M303 question asks you to perform calculations in a finite field. The question is based on concepts developed in Chapter 18.

Consider the finite field $F = \mathbb{Z}_2(\omega)$ where $\omega^4 = \omega + 1$; the order of F is $2^4 = 16$. By Proposition 1.7 of Chapter 18, every element of F can be written in a unique way in the form $a + b\omega + c\omega^2 + d\omega^3$ for some $a, b, c, d \in \mathbb{Z}_2$.

- (a) Show that for every element $z \in F$ we have $z + z = 0$. [1]
- (b) In this part of the question we follow the second method of Worked Exercises 3.10(b) and 3.11(c) of Chapter 18 and calculate the multiplicative inverse of the element $1 + \omega^2$ in F . We will find a, b, c and d so that $(1 + \omega^2)(a + b\omega + c\omega^2 + d\omega^3) = 1$.
- (i) Assume that $(1 + \omega^2)(a + b\omega + c\omega^2 + d\omega^3) = 1$.
Using the fact that $\omega^4 = 1 + \omega$ and $\omega^5 = \omega + \omega^2$; show that
- $$(a + c) + (b + c + d)\omega + (a + c + d)\omega^2 + (b + d)\omega^3 = 1. \quad [2]$$
- (ii) Hence calculate the values of a, b, c and d to find the multiplicative inverse of $1 + \omega^2$. [2]
- (c) Show that $\omega^3 \neq 1$ and $\omega^5 \neq 1$; hence, show that ω is a primitive element of F . [4]

Question 5 – 8 marks

This M303 question revises group theory.

For each of the following statements, write down whether it is true or false. Very briefly justify your answers (you do not have to provide detailed or formal proofs). The first two parts are done for you as examples.

- (a) Every dicyclic group, $\text{Dic}_n = \langle a, b \mid a^4 = e, a^2 = b^n, aba^3b = e \rangle$ has a subgroup of order 2.

Example solution

True. Reason: The element a^2 generates a subgroup of order 2.

- (b) Every abelian group is cyclic.

Example solution

False. The non-cyclic, abelian group $\mathbb{Z}_3 \times \mathbb{Z}_3$ is a counterexample.

- (c) Let $G \cong \mathbb{Z}_{10} \times \mathbb{Z}_p$ where p is an odd prime, with $p > 5$. Then the order of the element $(2, 2)$ is $5p$. [2]
- (d) If a group G has a normal abelian subgroup, H , such that G/H is abelian, then G is abelian. [2]
- (e) Let $G \cong \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_7$. Then G is the unique non-cyclic abelian group of order 56. [2]
- (f) Let $G \cong \mathbb{Z}_2 \times D_7$ where D_7 is the Dihedral group given by $\langle r, s \mid r^7 = s^2 = e, sr = r^6s \rangle$. Then none of the Sylow 2-subgroups of G are cyclic. [2]

Question 6 A— encryption – 10 marks

This M303 question tests your knowledge of encryption and decryption using the RSA method. The numbers involved are deliberately chosen to be small enough to focus on your understanding without excessive calculations.

Alice and Bob decide to use an RSA cryptosystem with public key $(299, 35)$ for communication.

- (a) Alice wants to send Bob the message $m = 143$. Determine Alice's ciphertext c . [4]
- (b) Determine Bob's private key. [2]
- (c) Carry out Bob's decryption of Alice's ciphertext c , and compare his result with Alice's message. [4]

Question 6 B— constructibility – 10 marks

This M303 question is designed to test your understanding of the (im)possibility of construction using ruler and compasses. It builds on the material of Chapter 19.

Duplication of a cube and quadrature of a circle are two classical problems of Greek geometry. Let us consider some related problems, namely the problems of ‘quadrupling a ball’ and ‘polygonalizing a circle’. Quadrupling a ball means constructing a ball of four times the volume of a given ball, and polygonalizing a circle means constructing a regular polygon with a finite number of sides with the same circumference as a given circle; in both cases, ‘constructing’ refers to using ruler and compasses only. We may also assume that the given ball and the given circle have unit radius.

- (a) Following the steps below, show that a unit ball cannot be quadrupled by a construction that only uses ruler and compasses.
 - (i) Show that if a unit ball can be quadrupled using ruler and compasses only, then $u = \sqrt[3]{4}$ is a constructible number. [2]
 - (ii) Find the minimal polynomial for u over $\mathbb{Q}$, and apply Corollary 2.8 of Chapter 19. [3]
- (b) Following the steps below, show that a unit circle cannot be polygonalized by a construction using only ruler and compasses.
 - (i) Show that if a unit circle can be polygonalized — that is, if a regular polygon with $n \geq 3$ sides with the same circumference as the circle can be constructed but with the help of ruler and compasses only — then $v = 2\pi/n$ is a constructible number. [2]
 - (ii) Show that v is transcendental over $\mathbb{Q}$, and apply Theorem 2.7 of Chapter 19. [3]

Question 7 – 3 marks

This M303 question can be done at any point during your study of Book E.

An important part of being a mathematician is being able to communicate mathematics using mathematical notation in different settings. For this TMA, we would like you to post something in the Book E forum that is about the mathematics in M303 and contains mathematical notation. There is a forum post explaining how to use mathematical notation in the Book E forum.

To earn full marks, provide either a screenshot of your forum post or directions as to where to find it (giving thread title, time and date). (If you are not in a position to provide such evidence, or if you cannot use the forum for any reason, contact your tutor for advice.) [3]

Part B Formative

Question 8 – 10 marks

This M303 question is designed to help you to practise techniques relevant to Question 1 in TMA 06. It concerns material covered in Chapter 21 of Book F.

Let d be the Euclidean metric on $\mathbb{R}^2$ given by

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

for $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$.

Consider the following four sets in $\mathbb{R}^2$:

$$W = \{(x, y) \in \mathbb{R}^2 : y < \pi\} \cup \{(x, y) \in \mathbb{R}^2 : y > \pi \text{ and } y \leq 7\},$$

$$X = \{(x, y) \in \mathbb{R}^2 : x = y\} \cup \{(x, 3) \in \mathbb{R}^2 : x \in \mathbb{R}\},$$

$$Y = \{(q, \pi) \in \mathbb{R}^2 : q \in \mathbb{Q}\},$$

$$Z = \{(x, y) \in \mathbb{R}^2 : x^2 + (y - \pi)^2 = \pi^2\} \cup \{(x, y) \in \mathbb{R}^2 : x^2 + (y + \pi)^2 < \pi^2\}.$$

For each set, do the following.

- (a) On a separate diagram, sketch the set, by hand or otherwise. (If you are unsure how to represent something on a diagram, or are unable to sketch it, then describe what you mean in words. You are also free to use a combination of sketching and writing.) [4]
- (b) Write down whether the set is d -connected or d -disconnected. If the set is d -disconnected, then write down a d -disconnection. [6]

Hint: when using the Euclidean metric, it is always worth guessing whether a set is disconnected and how it disconnects if it is disconnected.

Question 9 – 10 marks

This M303 question will test your calculation skills in rings of the form $\mathbb{Z}[\sqrt{-n}]$. It involves a number of concepts from Chapter 17 and also the concepts of prime and irreducible elements of a ring, introduced and studied earlier in Section 2, and specifically in Subsection 2.2, of Chapter 12.

Let $n \geq 3$ be an odd positive integer which is not a perfect square, and let $R = \mathbb{Z}[\sqrt{-n}]$, with multiplicative norm $N(r + s\sqrt{-n}) = r^2 + ns^2$ for every element $r + s\sqrt{-n} \in R$. You may assume that every element of R can be written uniquely as $a + b\sqrt{-n}$ where $a, b \in \mathbb{Z}$. Let $p \in \mathbb{Z}$, $p \geq 2$, be a prime in $\mathbb{Z}$ such that p divides $n + 1$.

- (a) By finding distinct factorisations of $n + 1$ in R , show that p is not a prime in R . [3]
Hint: See the solution to Exercise 2.13(b) of Chapter 17.
- (b) Show that there is no element $r + s\sqrt{-n} \in R$, $r, s \in \mathbb{Z}$, with norm equal to p . [3]
Hint: Determine first which of the two numbers p, n is smaller than the other.
- (c) With the help of the result of part (b) or otherwise, show that p is an irreducible element of R . [4]

Question 10 – 8 marks

This M303 question is a revision question on the material of Chapter 18, Section 2 about fields and their extensions. It is aimed at testing and developing your skills in handling extensions of the field $\mathbb{Q}$ of rational numbers by adjoining algebraic elements.

- (a) Determine the minimal polynomials for $\sqrt[3]{7}$ and $\sqrt[4]{7}$ over $\mathbb{Q}$ and determine the degrees $[\mathbb{Q}(\sqrt[3]{7}) : \mathbb{Q}]$ and $[\mathbb{Q}(\sqrt[4]{7}) : \mathbb{Q}]$. [3]
- (b) By considering common factors of the degrees found in part (a), or otherwise, show that $\sqrt[3]{7} \notin \mathbb{Q}(\sqrt[4]{7})$. [2]
- (c) Hence, determine the degree $[\mathbb{Q}(\sqrt[3]{7}, \sqrt[4]{7}) : \mathbb{Q}]$. [1]
- (d) Show that $\mathbb{Q}(\sqrt[3]{7}, \sqrt[4]{7}) = \mathbb{Q}(\sqrt[3]{7} \cdot \sqrt[4]{7}) = \mathbb{Q}(\sqrt[12]{7})$. [2]

Hint: Express the various roots of 7 as powers of $7^{\frac{1}{12}}$.

Question 11 – 11 marks

This M303 question is aimed at testing and developing your skills in handling extensions of fields by adjoining algebraic elements.

Consider the finite field $F = \mathbb{Z}_5(\omega)$ where $\omega^2 = -3 - 2\omega = 2 + 3\omega$; the order of F is $5^2 = 25$. Proposition 1.7 of Chapter 18 tells us that every element of F can be written uniquely in the form $a + b\omega$ for some $a, b \in \mathbb{Z}_5$.

By Theorem 3.3 of Chapter 18, the multiplicative group of F is cyclic, of order $5^2 - 1 = 24$. In particular, if ξ is a primitive element of F (Definition 3.4 of Chapter 18), then every non-zero element of F is equal to ξ^i for some i such that $0 \leq i \leq 23$.

- (a) Determine ω^3 and ω^6 in the form $a + b\omega$ for suitable $a, b \in \mathbb{Z}_5$. [2]
- (b) Show that ω is a primitive element of $\mathbb{Z}_5(\omega)$. [3]
- (c) By one of the methods explained in Worked Exercises 3.10 and 3.11 of Chapter 18 or otherwise, determine the multiplicative inverse ω^{-1} of ω in the form $\omega^{-1} = a + b\omega$ for suitable $a, b \in \mathbb{Z}_5$. [2]
- (d) With the help of Corollary 3.5 of Chapter 18 determine all primitive elements of $\mathbb{Z}_5(\omega)$ in the form $a + b\omega$ for suitable $a, b \in \mathbb{Z}_5$. [4]

Question 12 A— constructibility – 11 marks

This M303 question is designed to test your understanding of (im)possibility of construction with the help of ruler and compasses. It builds on the material of Chapter 19.

- (a) Assume that you have a ruler with marks at 0, 1 and $\sqrt[6]{2}$. Determine if it is possible to square a circle of radius $\sqrt{2}$ with such a ruler and compasses. [4]
- (b) Answer a similar question as in part (a) but for a ruler with marks at 0, 1 and π^2 . [3]
- (c) Determine if it is possible to duplicate a unit cube with compasses and a ruler with marks at 0, 1 and $\sqrt[3]{4}$. [2]
- (d) *Triplicating* a unit cube means constructing a side of a cube of volume 3. Determine if it is possible to triplicate a unit cube with compasses and a ruler with marks at 0, 1 and $\sqrt[6]{3}$. [2]

Question 12 B— encryption – 11 marks

This M303 question asks you to demonstrate your knowledge of encryption and decryption in cryptosystems based on elliptic curves over finite fields, covered in Chapter 20. The numbers involved are deliberately chosen to be small enough to focus on your understanding without excessive calculations.

Alice and Bob want to communicate using the Diffie-Hellman-ElGamal system based on the elliptic curve E over GF_{11} given by the equation $y^2 = x^3 + 2x + 5$, with the point $P = (0, 4)$. Bob decided to have $s_B = 2$ as his private key. By experimenting in the group $(E, +)$ Alice found out that $5P = (4, 0)$.

- (a) Using the above information, show that the order of P in the group $(E, +)$ is equal to 10. [2]
Hint: Determine first the order of the point $(4, 0)$ by working out its inverse in $(E, +)$.
 - (b) With the help of (a) and the result of Theorem 2.20 of Chapter 20 determine the order of $(E, +)$ and show that this group is cyclic, without listing any of the remaining elements. [3]
 - (c) Determine the point $P_B = s_BP$. [2]
 - (d) Alice wants to encrypt the message $M = (0, 7) = -P \in E$, choosing a random integer $k = 3$. Determine the points M_1 and M_2 Alice sends to Bob. [3]
 - (e) Describe Bob's decryption process of Alice's messages M_1 and M_2 . [1]
Hint for : Calculations in parts (c) and (d) will simplify if you realise that, by (a) and (b), every element of $(E, +)$ is a multiple of P .
-